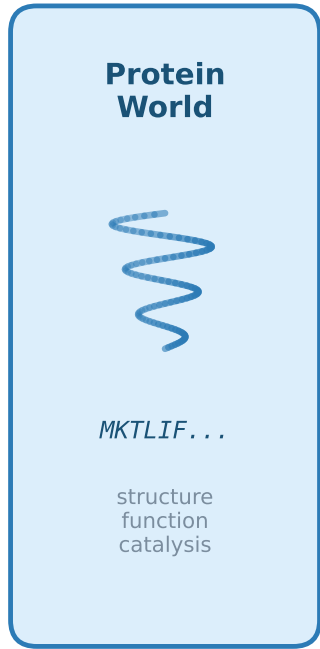
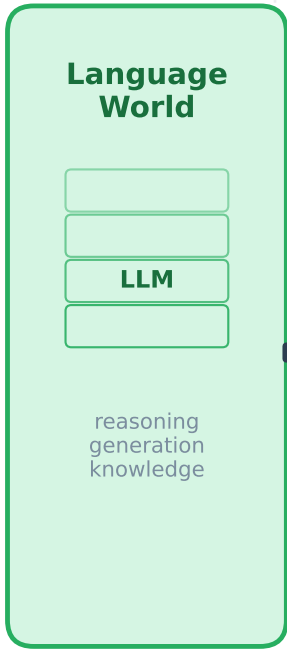


1 Question

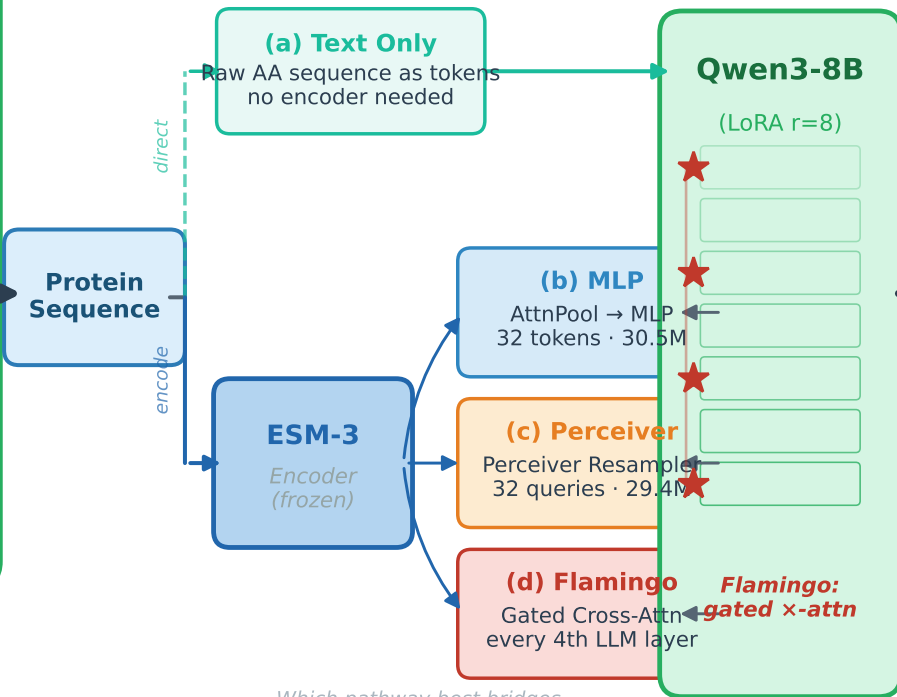


?



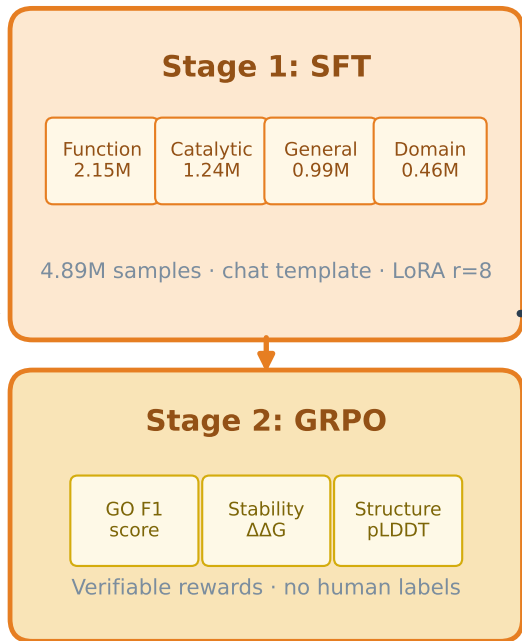
Can LLMs understand proteins
via structural embeddings?

2 Approach: Four Pathways

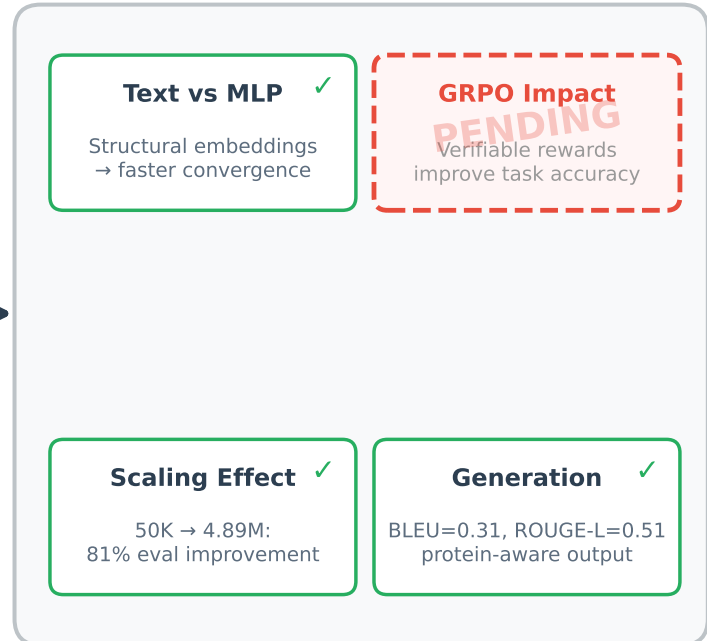


Which pathway best bridges
protein structure → language?

3 Training Pipeline



4 Key Results



Post-Training Protein LLM: Bridging Structural Biology and Language Models